

Probability

Chapter 1

Introduction

Richard Redner

August 21, 2026

We often model the world with **deterministic models**.

Linear algebra is used to model linear relationships between variables and to solve or approximately systems of equations.

Individual and systems of **ordinary differential equations (ODE's)** model changing relationships between variable when they depend on each other and a single variable, often time. For example, ODE's are often used for population modeling and the rate of spread of infectious diseases.

Partial differential equations (PDE's) arise naturally when the dependent variable depends on more than one variable and leads to the heat equation, the Navier Stokes equation for fluid dynamics and the wave equation for seismic modeling just to give a few examples.

Complementary to this are the disciplines of **Statistics and Probability**.

Statistics of course uses probabilistic models and measurements to draw inferences about the real world where outcomes cannot be known prior to making an observation.

Typical topics are model building, parameter estimation and inference. Statistics is intimately tied to probability.

Probability is the study of random variables and random processes. Given a probabilistic model, we can then determine the probability of future events and answer important and interesting questions.

Central to the theory of probability is the limiting behaviour of infinite sequences of events. The Strong Law of Large Numbers is a prime example.

Example 1. Given that I have drawn a Jack of Spades, what is the probability that the next card will be another Jack? What is the probability the next card will be a Spade?

This is an example of a conditional probability.

Example 2. Given a random graph, what is the probability that it is connected?

Example 3. Given a queue with two servers, what is the average waiting time? How should we model the queue so that it applies to real problems that we are interested in?

Example 4. Suppose that have a random walk in the plane that starts at the origin and takes unit steps randomly in the x and y direction.

What is the probability that the random walk will eventually return to origin?

What is the probability that the random walk will return to the origin infinitely often?

What is the probability that the random walk will move to $(5, 7)$ infinitely often?

Does the answer change if we are moving in one or in three dimensions?

Learn to develop and apply probability models to a variety of situations and to use those models to answer fundamental questions about the model.

Understand the fundamental theorems of probability theory; for example Baye's theorem, the strong law of large numbers and the central limit theorem.

Learn to effectively use some of the principle ideas of probability theory including the probability of an event or outcome, expectation, variance, conditional expectation, moment generating function, and probability density functions in one and more dimensions.

This semester we will investigate two particular areas of emphasis: Markov chains and queuing theory.

We will see that in both these applications and many others, the notion of a conditional expect will be a critical tool.

I will periodically use Mathematica to demonstrate examples and for computation. You will also need to use Mathematica from time to time as a tool for basic, but perhaps complex calculations, and for simulating queues and Markov Chains.

Every Notebook that I use in class will be made available to you on Harvey.

I do not anticipate using Matlab or R and certainly not Python, in this course.

Def. (Sample Space) Given an experiment, the sample space is the set of all possible outcomes from a random experiment. We will frequently use S to denote the sample space.

Def. (Event) Given a sample space S , an event is a subset of S . We will frequently use E and F to denote events. Note that we are using capital letters to denote sets.

Example 5. Consider the random experiment of rolling a six sided die and recording the outcome. Then

$$S = \{1, 2, 3, 4, 5, 6\}.$$

One possible event is that the outcome is an odd number. In this case $E = \{1, 3, 5\}$.

Example 6. The possible outcomes from measuring the battery life in months of a battery is a non-negative real number. And so

$$S = [0, \infty).$$

One possible event is that the battery lasts more than two years, in which case $E = \{t \geq 24\}$.

Example 7. If the experiment is to roll two dice, then the set of outcomes is

$$S = \{(i, j) | 1 \leq i, j \leq 6\},$$

which has 36 elements.

One possible event is the set E where the two die are different. A second possible event is, say F , where the sum of the two die is less than 6.

Question. What is the probability of E , that is the probability that two die are different?

Answer. There are 36 possible outcomes, each of which is equally likely. So the probability of each individual outcome is $\frac{1}{36}$.

Of these 36 possible outcomes, in $36 - 6 = 30$ cases, the two die are different. So the probability that the two die are distinct is $\frac{30}{36} = \frac{5}{6}$.

Question. What is $P(E \cap F)$, the probability that the two die are different and their sum is less than 6? **Note:** The outcome $(1, 2)$ is distinct from the outcome $(2, 1)$.

Let's take a more formal approach and define what we mean by a probability function.

Def. Let S be a set and for each $E \subset S$ and define a number $P(E)$ with the following properties.

(i.) For all E , $0 \leq P(E) \leq 1$.

(ii.) $P(S) = 1$.

(iii.) For any set of events $E_1, E_2, \dots$ that are mutually exclusive, which is to say that $E_i \cap E_j = \emptyset$, for $i \neq j$, we have that

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

Any such function is called a probability function.

iii.
$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

Item iii. is referred to as countable additivity and implied in the notation is that this also holds for a finite union of mutually exclusive (disjoint) sets.

As a consequence of iii., we have that $P(\emptyset) = 0$. This follows by observing that

$$P(\emptyset) = P(\emptyset \cup \emptyset) = P(\emptyset) + P(\emptyset) = 2P(\emptyset).$$

Furthermore, if $E \subset S$ and E^c is the complement of E in S , then

$$1 = P(S) = P(E \cup E^c) = P(E) + P(E^c)$$

and so

$$P(E^c) = 1 - P(E).$$

Let A and B be sets. Then

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

and

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

As an extension, if we let B_i be sets for $i = 1, 2, \dots$, then

$$A \cup \left(\bigcap_{i=1}^n B_i \right) = \bigcap_{i=1}^n (A \cup B_i)$$

and

$$A \cup \left(\bigcap_{i=1}^{\infty} B_i \right) = \bigcap_{i=1}^{\infty} (A \cup B_i).$$

Def. Let S be the set of all possible outcomes and let A be an event. Define

$$A^c = \{x \in S | x \notin A\}.$$

Theorem. (DeMorgan's Laws) For any sets A and B ,

$$(A \cup B)^c = A^c \cap B^c$$

and

$$(A \cap B)^c = A^c \cup B^c.$$

And if A_i are events for $i = 1, 2, \dots$, then

$$\left(\bigcup_{i=1}^n A_i \right)^c = \bigcap_{i=1}^n A_i^c$$

and

$$\left(\bigcup_{i=1}^{\infty} A_i \right)^c = \bigcap_{i=1}^{\infty} A_i^c.$$

Let A and B be sets.

Def. $A - B = A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$

We note that $A \cup B = A \setminus B \cup B$ and that $A \setminus B \cap B = \emptyset$.

Suppose that $B \subset A$. Then

$$P(A) = P(A \setminus B \cup B) = P(A \setminus B) + P(B) \geq P(B)$$

and so $P(A) \geq P(B)$.

If we let E and F be events, is it true that $P(E \cup F) = P(E) + P(F)$?

No, not unless $P(E \cap F) = 0$.

Question. What is $P(E \cup F)$?

Heuristically we are double counting the points in $E \cup F$. So the formula must be

$$P(E \cup F) = P(E) + P(F) - P(EF),$$

where we have used $EF = E \cap F$.

Homework. Prove this using only the axioms. You may need

$$E \cup F = (E - F) \cup (F - E) \cup (E \cap F).$$

Let's extend this idea to the union of three events, say E , F and G .

To determine $P(E \cup F \cup G)$ use the previous fact as follows:

$$\begin{aligned} P(E \cup F \cup G) &= P((E \cup F) \cup G) \\ &= P(E \cup F) + P(G) - P((E \cup F) \cap G) \\ &= P(E \cup F) + P(G) - P(EG \cup FG) \\ &= P(E) + P(F) - P(EF) + P(G) - P(EG) - P(FG) + P(EFG) \\ &= P(E) + P(F) + P(G) - P(EF) - P(EG) - P(FG) + P(EFG) \end{aligned}$$

Do we see a pattern?

Theorem. Let $E_1, E_2, \dots, E_n$ be n events.

Then

$$\begin{aligned} P(E_1 \cup E_2 \cup \dots \cup E_n) &= \sum_i P(E_i) - \sum_{i < j} P(E_i E_j) \\ &\quad + \sum_{i < j < k} P(E_i E_j E_k) \\ &\quad - \sum_{i < j < k < l} P(E_i E_j E_k E_l) \\ &\quad + \dots + (-1)^{n+1} P(E_1 E_2 \dots E_n). \end{aligned}$$

Example 8. Suppose that we roll 2 dice. What is the probability that the first die is 2 given that the sum is 6?

Solution. Define three sets. The set of all possible outcomes is

$$S = \{(i, j), 1 \leq i, j \leq 6\}.$$

The set where the first die is 2 is

$$E = \{(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6)\} \text{ and } P(E) = \frac{6}{36}.$$

The event that the sum of the two die is 6 is

$$F = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\} \text{ and } P(F) = \frac{5}{36}.$$

Without knowing the the sum is 6, the probability that the first die is 2 is $\frac{1}{6}$.

But if we know that the sum is six, then $P(E|F)$ (read the probability of E given F) is

$$P(E|F) = \frac{1}{5}.$$

Def. (Conditional Probability) Let E and F be subsets of S . If $P(F) \neq 0$, we define the conditional probability of E given F by

$$P(E|F) = \frac{P(EF)}{P(F)}.$$

This implies of course that $P(EF) = P(E|F) P(F)$.

Example 9. Suppose that we flip a fair coin three times and record the outcomes. Then the sample space S is

TTT THT HTT HHT

TTH THH HTH HHH

Question. What is the probability that we have at least two T's given that the first flip is a T?

Answer. We define $E = \{TTT, THT, HTT, TTH\}$, which is the set of outcomes with at least 2 T's.

We define $F = \{TTT, TTH, THT, THH\}$, which is the set of outcomes that starts with T and note that $E \cap F = EF = \{TTT, TTH, THT\}$

$$\text{Then } P(E|F) = \frac{P(EF)}{P(F)} = \frac{3/8}{4/8} = \frac{3}{4}.$$

Intuitively we should conclude that if E is independent of F , then

$$P(E|F) = P(E).$$

Or

$$P(E) = P(E|F) = \frac{P(EF)}{P(F)}$$

so that $P(EF) = P(E)P(F)$.

So if E is independent of F , then

$$P(EF) = P(E)P(F).$$

But then if E is independent of F , the

$$P(EF) = P(E)P(F)$$

and so

$$P(F|E) = \frac{P(EF)}{P(E)} = \frac{P(E)P(F)}{P(E)} = P(F)$$

and so E is independent of F implies that F is independent of E .

Def. (Independence) Let E and F be subsets of S . We say that E and F are independent iff

$$P(EF) = P(E)P(F).$$

Question. Suppose that we flip a fair coin 4 times and record the results. Let E be the event that the last coin is an H and let F be the event that the first three coins are H. Show that E and F are independent events.

Answer. S is composed of 16 sequences.

Let E = all sequences in S where the last coin comes up H. Then $P(E) = \frac{8}{16}$.

Let F = all sequences in S that start with HHH. Then $P(F) = \frac{2}{16}$.

And EF has only one element so, $P(E)P(F) = \frac{8}{16} \frac{2}{16} = \frac{1}{16} = P(EF)$. So E and F are independent.

As a result we note that $P(E|F) = P(E)$. So that the probability that we have observed three H's in a row does not affect the probability that we observe of fourth H. This is obvious unless you don't believe that this is true.

Lemma. Let $E = \cup_{i=1}^n F_i$, where $F_1, F_2, \dots, F_n$ are mutually exclusive events. (If each F_i is non-empty this could also be called a partition of E .) Then

$$P(E) = \sum_{i=1}^n P(E|F_i)P(F_i).$$

Proof.

$$\begin{aligned} P(E) &= P(E \cap E) \\ &= P(E \cap (\cup_{i=1}^n F_i)) \\ &= P(\cup_{i=1}^n (E \cap F_i)) \\ &= \sum_{i=1}^n P(E \cap F_i) \\ &= \sum_{i=1}^n P(E|F_i)P(F_i). \end{aligned}$$

Theorem. (Baye's formula) Let $E = \cup_{i=1}^n F_i$, where $F_1, F_2, \dots, F_n$ are mutually exclusive events. Then

$$\begin{aligned} P(F_j|E) &= \frac{P(EF_j)}{P(E)} \\ &= \frac{P(E|F_j)P(F_j)}{\sum_{i=1}^n P(E|F_i)P(F_i)}. \end{aligned}$$

Proof. That essentially is the proof since

$$P(E) = \sum_{i=1}^n P(E|F_i)P(F_i)$$

and

$$P(E|F_j) = \frac{P(EF_j)}{P(F_j)}$$

and so

$$P(EF_j) = P(E|F_j)P(F_j).$$

There are many such useful formulas and important inequalities that will be usefull. In particular

Homework. Problem 1.8 (Bonferroni's inequality) and 1.10 (Boole's inequality).

Example 1.14 A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a “false positive” result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then with probability 0.01, the test result will imply he has the disease.) If 0.5 percent of the population actually have the disease, what is the probability a person has the disease given that his test result is positive?

Solution. Let D be the event that the tested person has the disease, and E the event that his test result is positive. The desired probability $P(D|E)$ is obtained by

$$\begin{aligned} P(D|E) &= \frac{P(DE)}{P(E)} = \frac{P(E|D)P(D)}{P(E|D)P(D) + P(E|D^c)P(D^c)} \\ &= \frac{(.95)(.005)}{(0.95)(0.005) + (0.01)(0.995)} \\ &= \frac{95}{294} \approx 0.323. \end{aligned}$$

So only 32 percent of those persons whose test results are positive actually have the disease. I will leave it to you to consider the social and psychological implications of these calculations.

This is a breif summary of Chapter 1. of our text, which you should be before the next class.

Hopefully, most of this is familiar, but please focus on any ideas that are not well understood.

The End of the Beginning